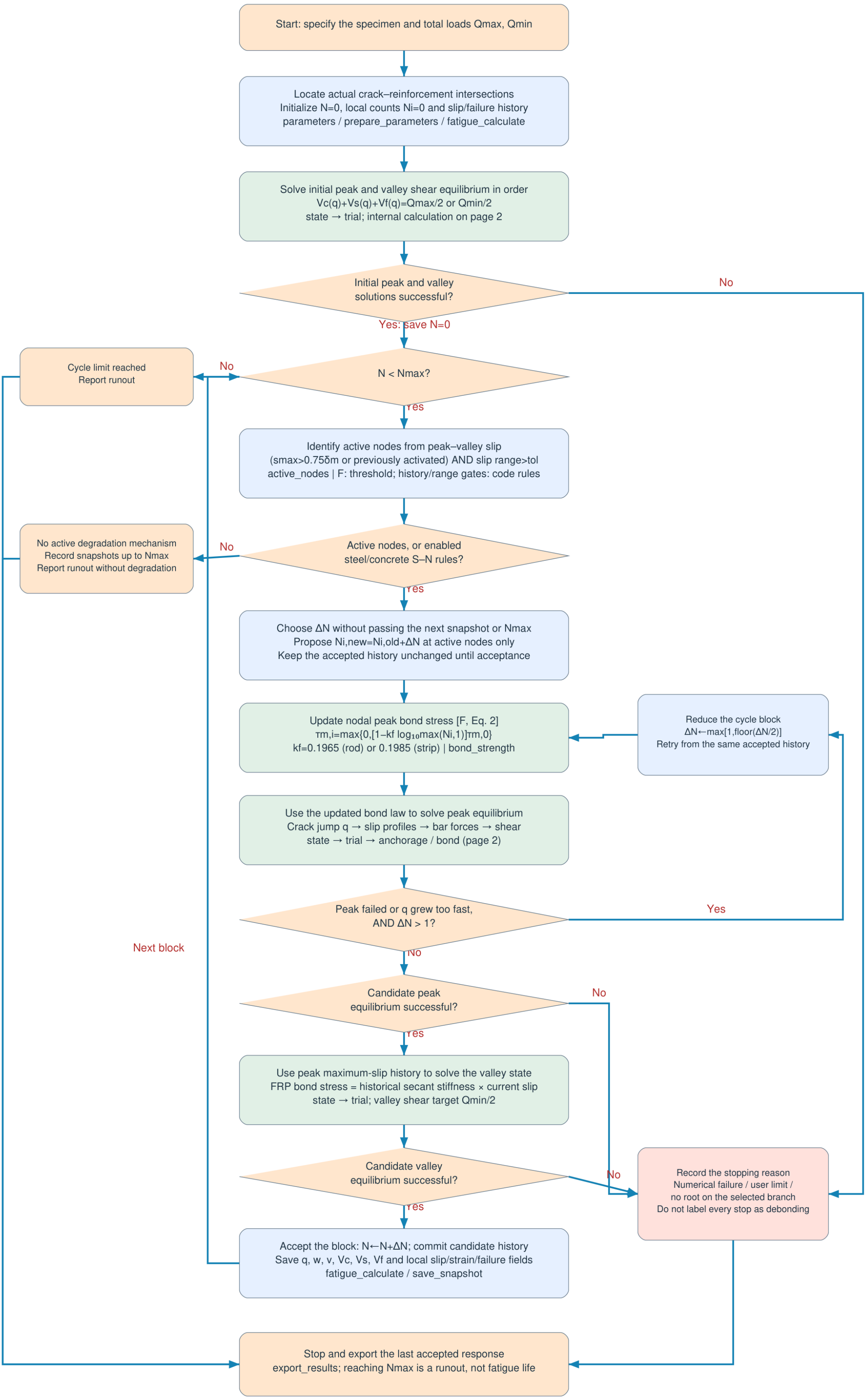
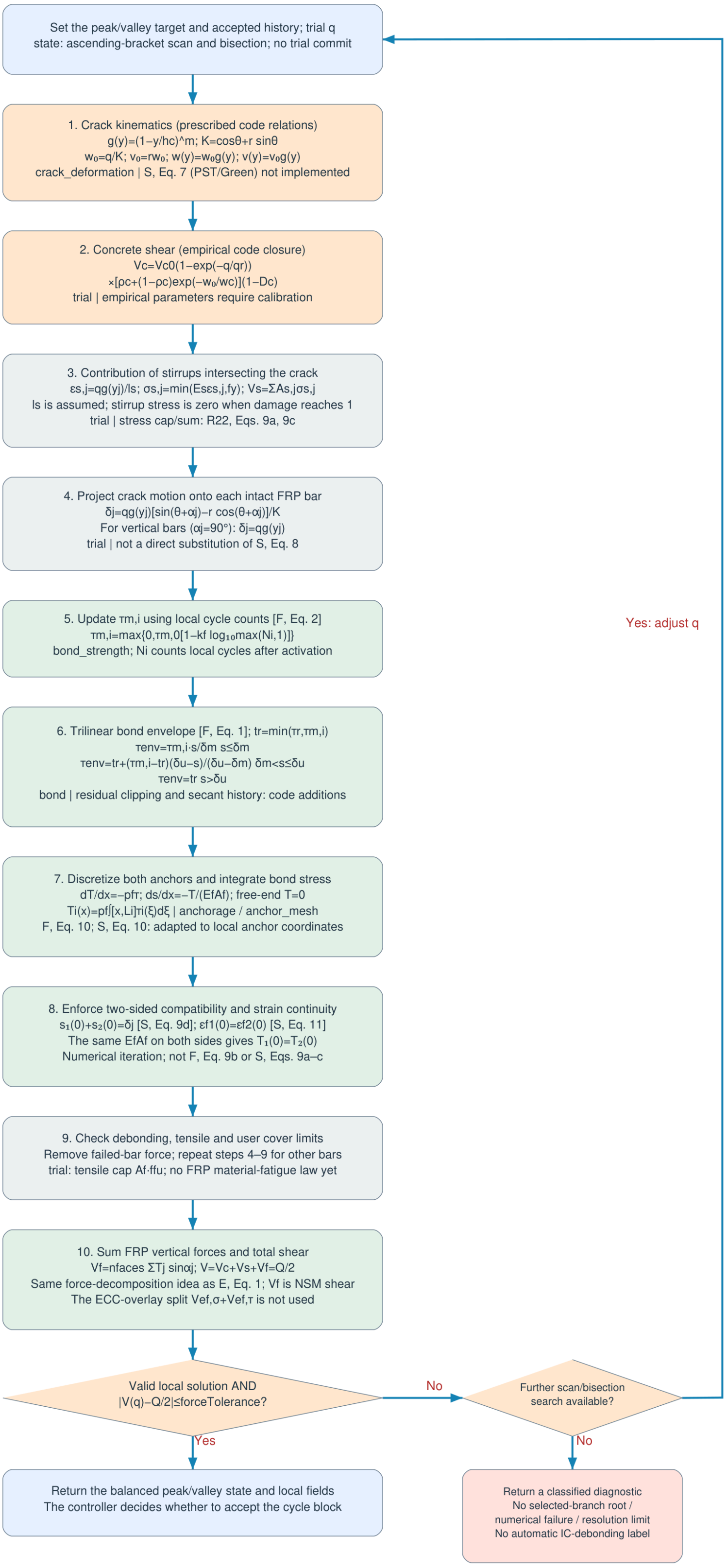
Formula-Mapped Shear Fatigue Main Loop



F: 2019 flexural-fatigue modeling paper; S: 2019 single-crack shear paper; E: 2024 ECC-FRP paper. Equation numbers need a paper prefix.

Formula-Mapped Shear Equilibrium Solver



F: DOI 10.1016/j.ijfatigue.2019.04.039; S: DOI suffix 0000953; E: CCENG-4395. Orange boxes include model assumptions.

Equation Scope and Current Implementation

How the supplied equations relate to the current shear code

Adopted: F, Eq. (1), trilinear envelope; F, Eq. (2), local fatigue degradation
bond.m / bond_strength.m; the 0.75δm threshold is used in active_nodes.m
The code adds zero-count protection, nonnegative peaks and a residual-stress rule.

Adapted principle: bond-stress integration in F, Eq. (10), and S, Eq. (10)
Replace half-span coordinates with local coordinates on each anchor: Ti(x)=pf∫[x,Li]τi dξ
S, Eqs. (9d) and (11), motivate two-sided slip compatibility and intersection strain continuity.

Not directly implemented: F, Eq. (9b), analytical matrix; S, Eqs. (9a–c), analytical slip fields
anchorage.m assembles bar/bond stiffness and uses fixed-point iteration with a Newton fallback.
The numerical system is not the paper matrix merely because both are written as matrix equations.

Not included: Ps/Pd from F, Eq. (11), and half-span Ls/Ld relations from F, Eq. (13)
These belong to the longitudinal flexural IC-debonding model; its P is not this model's total load Q.
The current code does not calculate critical loads or half-span damage-zone lengths using these equations.

Not included: longitudinally strengthened beam deflection from F, Eq. (14)
The eccentric longitudinal-reinforcement bending expression does not transfer directly to vertical NSM shear.
Here q is a vertical crack jump and w is crack opening; the paper's deflection w is a different quantity.

Adopted equilibrium idea: E, Eq. (1), V=Vc+Vs+Vef
Here V=Vc+Vs+Vf, where Vf sums the actual NSM bar-force vertical components.
E, Eq. (2), Vef,σ+Vef,τ, concerns the continuous ECC overlay and is not implemented for these NSM bars.

This diagram describes the current calculation; the MATLAB source remains unchanged.
Implementing F, Eqs. (9b), (11), (13), or the complete S crack model requires separate modeling and validation.
Equations not yet implemented are not presented as executed calculation steps.

R22: DOI suffix 0001251, intersecting reinforcement and shear-force decomposition. This update translates the diagrams only.